

Smart Canteen Ordering and Management System

Using Spring Boot and React

SOA Project Review-2

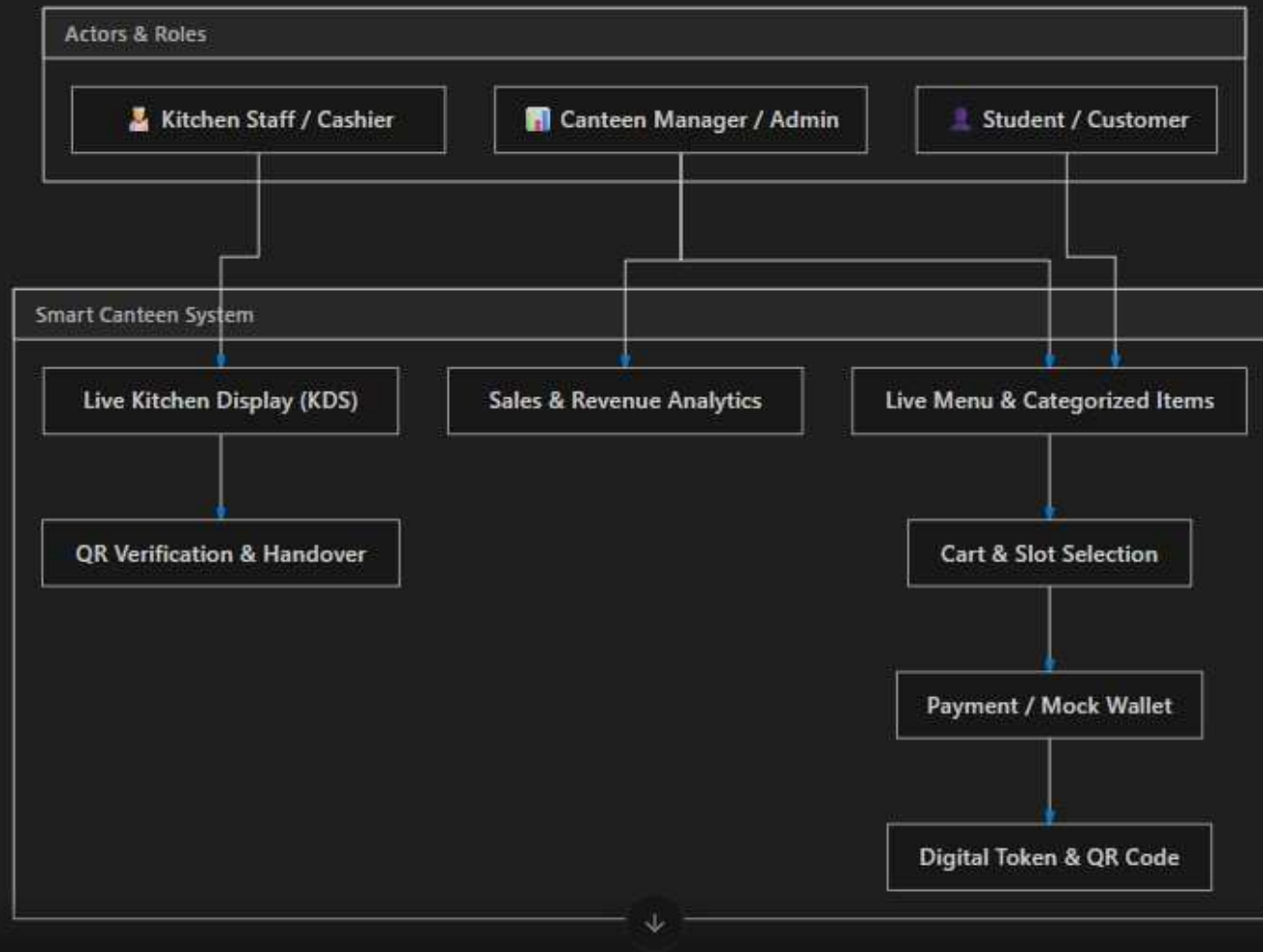
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Project Design – User Roles & Core Features

2. User Roles and Core Features



Smart Canteen Ordering and Management System

System Architecture



Student

Browse menu | Place order
Track order | QR pickup



Canteen Staff / Cashier

View orders | Update status
Manage menu & inventory



Admin

Manage users | Reports
System settings



API Gateway

Authentication | Routing | Load Balancing

SOA / Microservices



User Service

Manage profiles,
roles & access



Menu Service

Manage food items,
categories & availability



Order Service

Handle orders, cart,
payments & order status



QR Verification Service

Generate QR / Order ID
& verify pickup



Inventory Service

Track stock levels
& food availability



Notification Service

Alerts, order updates
& feedback



Database

MySQL | PostgreSQL | H2

Stores users, menu, orders,
order items, payments & inventory

Methodology

Requirement Analysis – Identify student, kitchen staff/cashier, and canteen manager/admin requirements for ordering, preparation, payment, pickup, and reporting.

Role-Based Design – Define separate access and responsibilities for Student/Customer, Kitchen Staff/Cashier, and Canteen Manager/Admin.

Menu & Order Workflow – Design categorized menu browsing, cart/slot selection, payment/mock wallet, token generation, and order-status flow.

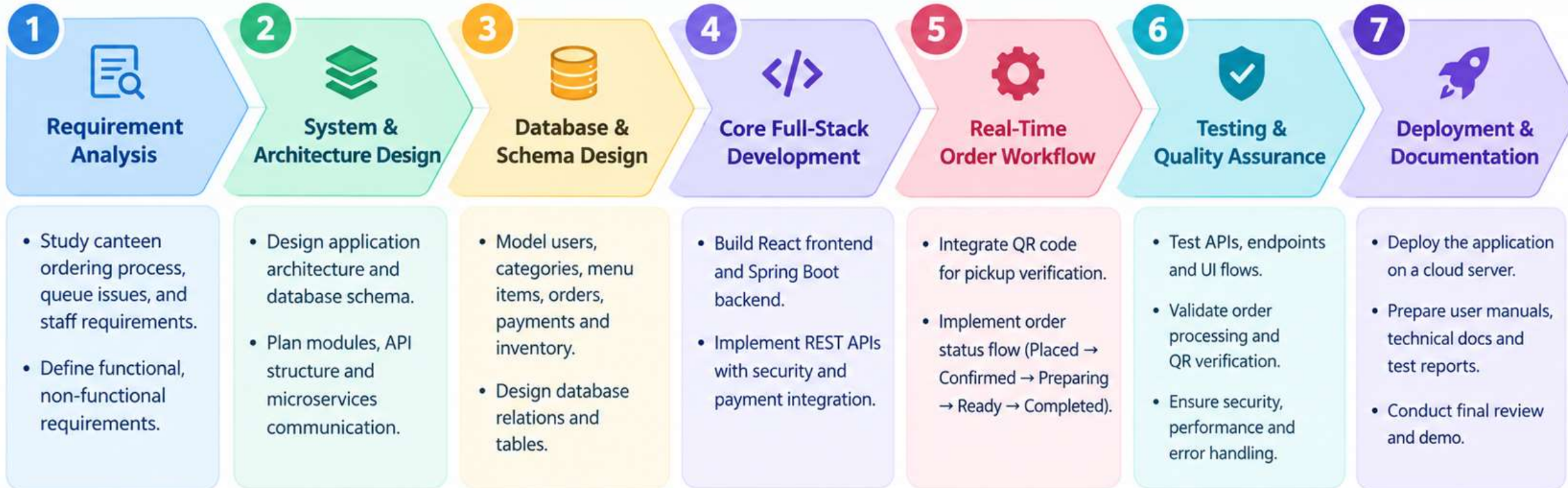
Real-Time Order Processing – Connect the student interface with the Live Kitchen Display (KDS) for instant order-status updates.

Pickup Verification – Use digital token and QR-code verification to support accurate order handover.

Integration & Testing – Integrate React, Spring Boot REST APIs, WebSocket/STOMP, security, and relational database components and test the complete workflow.

Methodology Adopted – 7-Phase Engineering Lifecycle

Smart Canteen Ordering and Management System



Technical Soundness

Frontend: React + Vite + Tailwind CSS provide a responsive, component-based interface for Student/Customer and Canteen Manager/Admin workflows.

Backend: Spring Boot 3.x with Java 17/21/25 provides REST APIs, dependency injection, business logic, and integration with the frontend.

Security: Spring Security + JWT supports stateless authentication and role-based access for ROLE_CUSTOMER, ROLE_STAFF, and ROLE_ADMIN.

Real-Time: Spring WebSocket + STOMP enables live order-status and kitchen updates without requiring manual refresh.

Database: MySQL / PostgreSQL, with H2 for testing, stores users, menu categories, menu items, orders, order items, and payment records.

Overall: The layered architecture connects the client, REST/WebSocket backend, persistence layer, and canteen workflows into one centralized system.

Technical Soundness – Production Technology Stack

Layer	Recommended Technology	Why?
Frontend	React (Vite) + Tailwind CSS	Fast build time, reactive UI state, and responsive mobile-first design.
Icons & QR	Lucide-React & qrcode.react	Clean modern icons and dynamic SVG/Canvas QR code generation.
Backend	Spring Boot 3.x (Java 17/21/25)	Industry standard, robust REST APIs, dependency injection, and security.
Security	Spring Security + JWT	Stateless authentication with role-based access control (<code>ROLE_CUSTOMER</code> , <code>ROLE_STAFF</code> , <code>ROLE_ADMIN</code>).
Real-Time	Spring WebSocket + STOMP	Instant notification to kitchen screen and live status updates to customer phones.
Database	MySQL / PostgreSQL (or H2 for testing)	Relational integrity for orders, items, tokens, and payments.

Implementation Modules

Module 1 – User & Role Management

Registration/login and role-based access for Student/Customer, Kitchen Staff/Cashier, and Canteen Manager/Admin.

Module 2 – Menu & Category Management

Maintain categorized food items, prices, descriptions, images, and availability.

Module 3 – Cart, Slot & Payment

Students select items, choose a pickup slot, place the order, and complete payment through the planned payment/mock-wallet flow.

Module 4 – Order Processing & KDS

Orders move through status stages and appear on the Live Kitchen Display for staff to prepare and update.

Module 5 – Digital Token & QR Handover

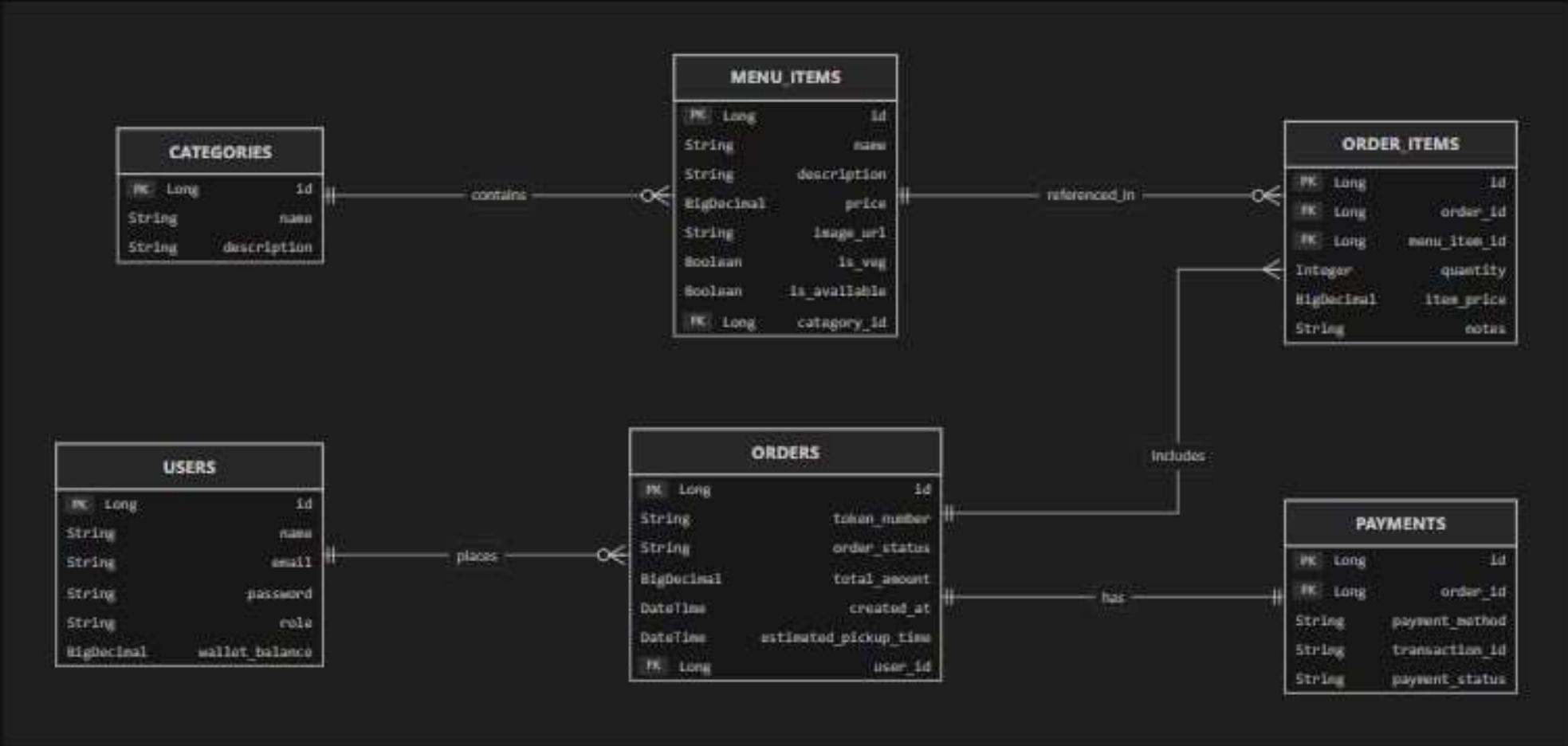
Generate a digital token/QR code and verify the order during pickup.

Module 6 – Sales & Revenue Analytics

Provide basic sales and revenue views for canteen management.

Database Schema – Entity Relationship

4. Database Schema (Entity-Relationship)



THANK YOU